THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

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Batch :- T13 Class :- T1 Roll. No :- 63

Introduction :-

- In modern software development, organizations need to deliver applications faster & more reliably. Traditional developments & operations worked separately, leading to delays, conflicts & failures. DevOps emerged as a solution to overcome these challenges by integrating development & operations team. It helps in improving collaboration, automation & continous delivery.

Problems in Traditional approach :-

- Earlier developers focused only on adding new features, while operations teams focused on system stability
  - ↳ Integration was slow & unpredictable
  - ↳ Bugs were found late in testing
  - ↳ Manual configuration caused errors
  - ↳ Different environment caused inconsistencies

What is DevOps ?

- DevOps is a combination of cultural philosophies, practices & tools that increases an organization's ability to deliver applications at high speeds.
- It focuses on Communication, Collaboration & Continous Integration
- DevOps enables organizations to release updates quickly & efficiently, helping them stay competitive

Principles of DevOps (CALMS Framework)

1. Culture :-
   ↳ encourages collaboration between teams
   ↳ shared responsibility for product success

2. Automation
   ↳ reduces manual work
   ↳ ensures consistencies & reliable processes

3. Lean
   ↳ focus on continous improvement
   ↳ eliminates wastes and inefficiencies

4. Measurement
   ↳ tracks performance metric like deployment time, failures
   ↳ helps in better decision making

5. Sharing
   ↳ promotes knowledge sharing across teams
   ↳ improves transparency & teamwork

DevOps Practices

1. Continous Development
   ↳ Planning, coding & version control of software

2. Continous Testing
   ↳ automated testing to ensure quality

3. Continous Integration
   ↳ code changes are automatically built & tested.

4. Continous Delivery & deployment
↳ faster & reliable software release

5. Infrastructure as code

6. Configuration Management

7. Microservice Architecture
↳ breaks application into small independent services

8. Continous monitoring
↳ tracks performance & detects issues early

9. Cloud based DevOps
↳ uses cloud platforms for scalability and automation

Roles & Responsibilities of DevOps Engineer

1. automating build, test and deployment process
2. managing CI/CD pipelines
3. monitoring system performance & reliability
4. handling infrastructure using tools and scripts
5. ensuring security and system stability
6. collaborating with developers & operations team
7. managing cloud platforms and containers

Benefits of DevOps

1. faster software delivery
2. improved collaboration
3. reduced errors and failures

4. better customer satisfaction
5. continous improvement of products

Conclusion

DevOps is a modern approach that combines development & operations to improve efficiency, speed. By adopting DevOps principles & practices, organizations can deliver reliable software continously and meet customer demands efficiently